Imports

```
# PyTorch
import torch
print(f"Is CUDA available? {torch.cuda.is_available()}")
print(f"GPU Name: {torch.cuda.get_device_name(0)}")
print(f"CUDA Version: {torch.version.cuda}")
import torchvision
import torchvision.transforms as transforms
import torch.nn as nn
import torch.nn.functional as nnFunc
import torch.amp as amp
# General
import numpy as np
import matplotlib.pyplot as plt
# Optimizer
import torch.optim as optim
# pretty print
from pprint import pprint
```

```
Is CUDA available? True
GPU Name: NVIDIA GeForce RTX 5050 Laptop GPU
CUDA Version: 13.0
```

Dataset

This code uses the CIFAR-10 dataset. (https://www.cs.toronto.edu/~kriz/cifar.html)

1. Transform setting

The original images are in PIL (0 to 255 for each pixel). First we convert it to a tensor format (0 to 1). Then to make the data to be zero centering, we make the range to be between -1 to 1. If we use the following equation for each data, we can make it zero centered: \begin{equation} output = \frac{input - mean}{std} \end{equation} This is being done by setting the `mean` and `standard deviation` to be 0.5.

```
transform = transforms.Compose([  transforms.RandomHorizontalFlip(),
                                  transforms.RandomCrop(32, padding=4),
                                  transforms.ToTensor(),
                                  transforms.Normalize((0.4914, 0.4822,
0.4465), (0.2023, 0.1994, 0.2010))
                                  ])
```

2. Making the training set

Now we load the actual dataset and make the training dataset.

1. Load CIFAR10 dataset (Download and prepare dataset)
    a. Put it in `./data_CIFAR10` directory.
    b. Download it
    c. Do the specified transformation on the dataset
2. Create the data loader (pipelining settings)
    a. Brake it to batches with size of 32
    b. Shuffle the deck in every epoch
    c. Use 4 number of workers to make the training faster

```
batchSize = 256
numberOfWorkers = 0

trainSet = torchvision.datasets.CIFAR10(root='./data_CIFAR10',
                                        train=True,
                                        download=True,
                                        transform = transform)

trainLoader = torch.utils.data.DataLoader(trainSet,
                                          batch_size=batchSize,
                                          shuffle=True,
                                          num_workers=numberOfWorkers,
                                          pin_memory=True,
                                          persistent_workers=False)
```

3. Making the test set

Now making the test set. The procedure is the same as making the training set with some minor modifications.

```
testSet = torchvision.datasets.CIFAR10(root='./data_CIFAR10',
                                       train=False,
                                       download=True,
                                       transform=transform)

testLoader = torch.utils.data.DataLoader(testSet,
                                         batch_size=batchSize,
                                         shuffle=False,
                                         num_workers=numberOfWorkers,
                                         pin_memory=True,
                                         persistent_workers=False)
viz_batchSize = 4
vizLoader = torch.utils.data.DataLoader(testSet,
batch_size=viz_batchSize, shuffle=True)
```

4. The classes

```
classes = ('plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog',
'horse', 'ship', 'truck')
pprint(classes)
```

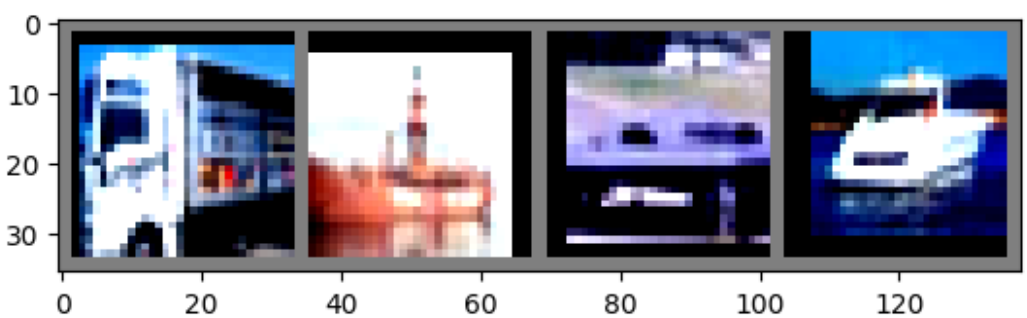
```
('plane', 'car', 'bird', 'cat', 'deer', 'dog', 'frog', 'horse',
'ship', 'truck')
```

DEUBG1

Showing some of the images from the dataset.

```
# functions to show an image
def imshow(img):
  img = img / 2 + 0.5 # unnormalize
  npimg = img.numpy()
  plt.imshow(np.transpose(npimg, (1, 2, 0)))
  plt.show()
# get some random training images
dataiter = iter(vizLoader)
images, labels = next(dataiter)
# show images
imshow(torchvision.utils.make_grid(images))
# print labels
pprint(' '.join(f'{classes[labels[j]]:5s}' for j in
range(viz_batchSize)))
```

```
Clipping input data to the valid range for imshow with RGB data
([0..1] for floats or [0..255] for integers). Got range [-
0.71453285..1.8768656].
```




```
'truck ship  plane ship '
```

5. Defining the CNN model

```
# Define the VGG-style convolutional neural network
class Net(nn.Module):
```

```python
    def __init__(self):
        super(Net, self).__init__()
        self.features = nn.Sequential(
            nn.Conv2d(3, 64, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(64, 64, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),
            nn.Conv2d(64, 128, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(128, 128, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),
            nn.Conv2d(128, 256, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(256, 256, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(256, 256, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),
            nn.Conv2d(256, 512, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.Conv2d(512, 512, kernel_size=3, padding=1),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(kernel_size=2, stride=2),
        )
        self.classifier = nn.Sequential(
            nn.Linear(512, 4096),
            nn.ReLU(inplace=True),
            nn.Dropout(),
            nn.Linear(4096, 4096),
            nn.ReLU(inplace=True),
            nn.Dropout(),
            nn.Linear(4096, 10),
        )

    def forward(self, x):
        x = self.features(x)
        x = x.view(x.size(0), -1)
```

```
        x = self.classifier(x)
        return x
```

6. Instansiating the CNN model

```
def init_weights(m):
    if isinstance(m, nn.Conv2d) or isinstance(m, nn.Linear):
        nn.init.kaiming_normal_(m.weight, mode='fan_out',
nonlinearity='relu')
        if m.bias is not None:
            nn.init.constant_(m.bias, 0)
net = Net()
net.apply(init_weights)
```

```
Net(
  (features): Sequential(
    (0): Conv2d(3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): ReLU(inplace=True)
    (2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (3): ReLU(inplace=True)
    (4): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (5): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (6): ReLU(inplace=True)
    (7): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (8): ReLU(inplace=True)
    (9): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (10): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (11): ReLU(inplace=True)
    (12): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (13): ReLU(inplace=True)
    (14): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (15): ReLU(inplace=True)
    (16): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (17): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (18): ReLU(inplace=True)
    (19): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (20): ReLU(inplace=True)
```

```
    (21): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (22): ReLU(inplace=True)
    (23): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (24): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (25): ReLU(inplace=True)
    (26): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (27): ReLU(inplace=True)
    (28): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (29): ReLU(inplace=True)
    (30): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Linear(in_features=512, out_features=4096, bias=True)
    (1): ReLU(inplace=True)
    (2): Dropout(p=0.5, inplace=False)
    (3): Linear(in_features=4096, out_features=4096, bias=True)
    (4): ReLU(inplace=True)
    (5): Dropout(p=0.5, inplace=False)
    (6): Linear(in_features=4096, out_features=10, bias=True)
  )
)
```

7. Move the model to GPU

```
device = torch.device('cuda:0' if torch.cuda.is_available() else
'cpu')
print(device)
```

```
cuda:0
```

```
net.to(device)
```

```
Net(
  (features): Sequential(
    (0): Conv2d(3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (1): ReLU(inplace=True)
    (2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1))
    (3): ReLU(inplace=True)
    (4): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (5): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
```

```
    (6): ReLU(inplace=True)
    (7): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (8): ReLU(inplace=True)
    (9): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (10): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (11): ReLU(inplace=True)
    (12): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (13): ReLU(inplace=True)
    (14): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (15): ReLU(inplace=True)
    (16): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (17): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (18): ReLU(inplace=True)
    (19): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (20): ReLU(inplace=True)
    (21): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (22): ReLU(inplace=True)
    (23): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (24): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (25): ReLU(inplace=True)
    (26): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (27): ReLU(inplace=True)
    (28): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1),
padding=(1, 1))
    (29): ReLU(inplace=True)
    (30): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Linear(in_features=512, out_features=4096, bias=True)
    (1): ReLU(inplace=True)
    (2): Dropout(p=0.5, inplace=False)
    (3): Linear(in_features=4096, out_features=4096, bias=True)
    (4): ReLU(inplace=True)
    (5): Dropout(p=0.5, inplace=False)
    (6): Linear(in_features=4096, out_features=10, bias=True)
```

```
  )
)
```

8. Degine loss function and optimizer

```
criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(net.parameters(), lr=0.01, momentum=0.9)

scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer,
mode='max', factor=0.1, patience=10)
```

9. Traing the network

```
scaler = amp.GradScaler('cuda')

epochs = 100
printFrequency = len(trainLoader) // 4

for epoch in range(epochs):
    net.train()
    running_loss = 0.0
    correct = 0
    total = 0

    # --- TRAINING PHASE ---
    for i, data in enumerate(trainLoader, 0):
        inputs, labels = data[0].to(device, non_blocking=True),
data[1].to(device, non_blocking=True)

        optimizer.zero_grad()

        with amp.autocast('cuda'):
            outputs = net(inputs)
            loss = criterion(outputs, labels)

        scaler.scale(loss).backward()
        scaler.unscale_(optimizer)
        torch.nn.utils.clip_grad_norm_(net.parameters(), max_norm=1.0)

        scaler.step(optimizer)
        scaler.update()

        _, predicted = torch.max(outputs.data, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()
        running_loss += loss.item()

        if i % printFrequency == (printFrequency - 1):
            accuracy = 100 * correct / total
```

```
            print(f'Epoch {epoch + 1} | Batch {i + 1}/{len(trainLoader)} | loss: {running_loss / printFrequency:.3f} | acc: {accuracy:.2f}%')
            running_loss, correct, total = 0.0, 0, 0

    # --- VALIDATION PHASE ---
    net.eval()
    val_correct = 0
    val_total = 0
    with torch.no_grad():
        for data in testLoader:
            images, labels = data[0].to(device), data[1].to(device)
            with amp.autocast('cuda'):
                outputs = net(images)
            _, predicted = torch.max(outputs.data, 1)
            val_total += labels.size(0)
            val_correct += (predicted == labels).sum().item()

    val_acc = 100 * val_correct / val_total
    current_lr = optimizer.param_groups[0]['lr']
    print(f'>> Epoch {epoch + 1} SUMMARY: Val Acc: {val_acc:.2f}% | Current LR: {current_lr}')

    # --- AUTOMATIC LR DROP ---
    scheduler.step(val_acc)

print('Finished Training')
```

```
Epoch 1 | Batch 49/196 | loss: 2.358 | acc: 12.24%
Epoch 1 | Batch 98/196 | loss: 2.075 | acc: 20.93%
Epoch 1 | Batch 147/196 | loss: 1.946 | acc: 25.57%
Epoch 1 | Batch 196/196 | loss: 1.860 | acc: 28.87%
>> Epoch 1 SUMMARY: Val Acc: 31.59% | Current LR: 0.01
Epoch 2 | Batch 49/196 | loss: 1.792 | acc: 31.61%
Epoch 2 | Batch 98/196 | loss: 1.732 | acc: 34.38%
Epoch 2 | Batch 147/196 | loss: 1.707 | acc: 35.49%
Epoch 2 | Batch 196/196 | loss: 1.670 | acc: 36.59%
>> Epoch 2 SUMMARY: Val Acc: 41.11% | Current LR: 0.01
Epoch 3 | Batch 49/196 | loss: 1.636 | acc: 38.55%
Epoch 3 | Batch 98/196 | loss: 1.592 | acc: 39.96%
Epoch 3 | Batch 147/196 | loss: 1.543 | acc: 42.11%
Epoch 3 | Batch 196/196 | loss: 1.545 | acc: 42.12%
>> Epoch 3 SUMMARY: Val Acc: 44.21% | Current LR: 0.01
Epoch 4 | Batch 49/196 | loss: 1.514 | acc: 43.81%
Epoch 4 | Batch 98/196 | loss: 1.501 | acc: 44.70%
Epoch 4 | Batch 147/196 | loss: 1.488 | acc: 45.65%
Epoch 4 | Batch 196/196 | loss: 1.457 | acc: 46.18%
>> Epoch 4 SUMMARY: Val Acc: 48.90% | Current LR: 0.01
Epoch 5 | Batch 49/196 | loss: 1.433 | acc: 46.87%
Epoch 5 | Batch 98/196 | loss: 1.424 | acc: 47.76%
```

```
Epoch 5 | Batch 147/196 | loss: 1.385 | acc: 49.43%
Epoch 5 | Batch 196/196 | loss: 1.367 | acc: 49.68%
>> Epoch 5 SUMMARY: Val Acc: 50.79% | Current LR: 0.01
Epoch 6 | Batch 49/196 | loss: 1.355 | acc: 50.45%
Epoch 6 | Batch 98/196 | loss: 1.338 | acc: 50.91%
Epoch 6 | Batch 147/196 | loss: 1.299 | acc: 52.07%
Epoch 6 | Batch 196/196 | loss: 1.299 | acc: 52.26%
>> Epoch 6 SUMMARY: Val Acc: 53.36% | Current LR: 0.01
Epoch 7 | Batch 49/196 | loss: 1.275 | acc: 52.52%
Epoch 7 | Batch 98/196 | loss: 1.265 | acc: 53.99%
Epoch 7 | Batch 147/196 | loss: 1.222 | acc: 55.74%
Epoch 7 | Batch 196/196 | loss: 1.216 | acc: 55.66%
>> Epoch 7 SUMMARY: Val Acc: 55.31% | Current LR: 0.01
Epoch 8 | Batch 49/196 | loss: 1.205 | acc: 56.75%
Epoch 8 | Batch 98/196 | loss: 1.204 | acc: 56.23%
Epoch 8 | Batch 147/196 | loss: 1.197 | acc: 56.63%
Epoch 8 | Batch 196/196 | loss: 1.174 | acc: 57.33%
>> Epoch 8 SUMMARY: Val Acc: 59.04% | Current LR: 0.01
Epoch 9 | Batch 49/196 | loss: 1.148 | acc: 58.39%
Epoch 9 | Batch 98/196 | loss: 1.133 | acc: 59.18%
Epoch 9 | Batch 147/196 | loss: 1.158 | acc: 58.11%
Epoch 9 | Batch 196/196 | loss: 1.116 | acc: 60.19%
>> Epoch 9 SUMMARY: Val Acc: 60.40% | Current LR: 0.01
Epoch 10 | Batch 49/196 | loss: 1.112 | acc: 59.72%
Epoch 10 | Batch 98/196 | loss: 1.084 | acc: 61.80%
Epoch 10 | Batch 147/196 | loss: 1.060 | acc: 62.16%
Epoch 10 | Batch 196/196 | loss: 1.070 | acc: 61.50%
>> Epoch 10 SUMMARY: Val Acc: 63.76% | Current LR: 0.01
Epoch 11 | Batch 49/196 | loss: 1.043 | acc: 62.76%
Epoch 11 | Batch 98/196 | loss: 1.061 | acc: 62.13%
Epoch 11 | Batch 147/196 | loss: 1.042 | acc: 62.64%
Epoch 11 | Batch 196/196 | loss: 0.994 | acc: 65.04%
>> Epoch 11 SUMMARY: Val Acc: 63.76% | Current LR: 0.01
Epoch 12 | Batch 49/196 | loss: 1.003 | acc: 64.33%
Epoch 12 | Batch 98/196 | loss: 0.996 | acc: 64.51%
Epoch 12 | Batch 147/196 | loss: 0.977 | acc: 65.62%
Epoch 12 | Batch 196/196 | loss: 0.992 | acc: 64.96%
>> Epoch 12 SUMMARY: Val Acc: 63.10% | Current LR: 0.01
Epoch 13 | Batch 49/196 | loss: 0.990 | acc: 64.87%
Epoch 13 | Batch 98/196 | loss: 0.951 | acc: 65.82%
Epoch 13 | Batch 147/196 | loss: 0.967 | acc: 65.86%
Epoch 13 | Batch 196/196 | loss: 0.930 | acc: 67.43%
>> Epoch 13 SUMMARY: Val Acc: 67.00% | Current LR: 0.01
Epoch 14 | Batch 49/196 | loss: 0.940 | acc: 66.63%
Epoch 14 | Batch 98/196 | loss: 0.927 | acc: 67.26%
Epoch 14 | Batch 147/196 | loss: 0.908 | acc: 67.72%
Epoch 14 | Batch 196/196 | loss: 0.905 | acc: 68.09%
>> Epoch 14 SUMMARY: Val Acc: 67.54% | Current LR: 0.01
Epoch 15 | Batch 49/196 | loss: 0.908 | acc: 68.28%
```

```
Epoch 15 | Batch 98/196 | loss: 0.896 | acc: 67.98%
Epoch 15 | Batch 147/196 | loss: 0.888 | acc: 68.34%
Epoch 15 | Batch 196/196 | loss: 0.900 | acc: 68.45%
>> Epoch 15 SUMMARY: Val Acc: 67.79% | Current LR: 0.01
Epoch 16 | Batch 49/196 | loss: 0.848 | acc: 70.08%
Epoch 16 | Batch 98/196 | loss: 0.884 | acc: 68.81%
Epoch 16 | Batch 147/196 | loss: 0.854 | acc: 69.86%
Epoch 16 | Batch 196/196 | loss: 0.875 | acc: 69.09%
>> Epoch 16 SUMMARY: Val Acc: 67.97% | Current LR: 0.01
Epoch 17 | Batch 49/196 | loss: 0.871 | acc: 69.56%
Epoch 17 | Batch 98/196 | loss: 0.862 | acc: 69.59%
Epoch 17 | Batch 147/196 | loss: 0.834 | acc: 70.45%
Epoch 17 | Batch 196/196 | loss: 0.821 | acc: 71.08%
>> Epoch 17 SUMMARY: Val Acc: 72.60% | Current LR: 0.01
Epoch 18 | Batch 49/196 | loss: 0.831 | acc: 70.68%
Epoch 18 | Batch 98/196 | loss: 0.816 | acc: 71.58%
Epoch 18 | Batch 147/196 | loss: 0.793 | acc: 72.11%
Epoch 18 | Batch 196/196 | loss: 0.817 | acc: 71.18%
>> Epoch 18 SUMMARY: Val Acc: 72.44% | Current LR: 0.01
Epoch 19 | Batch 49/196 | loss: 0.794 | acc: 71.90%
Epoch 19 | Batch 98/196 | loss: 0.810 | acc: 71.70%
Epoch 19 | Batch 147/196 | loss: 0.793 | acc: 72.24%
Epoch 19 | Batch 196/196 | loss: 0.761 | acc: 72.98%
>> Epoch 19 SUMMARY: Val Acc: 71.64% | Current LR: 0.01
Epoch 20 | Batch 49/196 | loss: 0.788 | acc: 71.99%
Epoch 20 | Batch 98/196 | loss: 0.797 | acc: 71.99%
Epoch 20 | Batch 147/196 | loss: 0.766 | acc: 73.46%
Epoch 20 | Batch 196/196 | loss: 0.765 | acc: 73.39%
>> Epoch 20 SUMMARY: Val Acc: 71.08% | Current LR: 0.01
Epoch 21 | Batch 49/196 | loss: 0.774 | acc: 73.03%
Epoch 21 | Batch 98/196 | loss: 0.754 | acc: 73.77%
Epoch 21 | Batch 147/196 | loss: 0.754 | acc: 73.25%
Epoch 21 | Batch 196/196 | loss: 0.752 | acc: 73.81%
>> Epoch 21 SUMMARY: Val Acc: 73.06% | Current LR: 0.01
Epoch 22 | Batch 49/196 | loss: 0.721 | acc: 74.76%
Epoch 22 | Batch 98/196 | loss: 0.726 | acc: 74.45%
Epoch 22 | Batch 147/196 | loss: 0.733 | acc: 74.63%
Epoch 22 | Batch 196/196 | loss: 0.732 | acc: 74.37%
>> Epoch 22 SUMMARY: Val Acc: 73.85% | Current LR: 0.01
Epoch 23 | Batch 49/196 | loss: 0.719 | acc: 75.00%
Epoch 23 | Batch 98/196 | loss: 0.717 | acc: 75.23%
Epoch 23 | Batch 147/196 | loss: 0.710 | acc: 74.99%
Epoch 23 | Batch 196/196 | loss: 0.714 | acc: 75.17%
>> Epoch 23 SUMMARY: Val Acc: 74.15% | Current LR: 0.01
Epoch 24 | Batch 49/196 | loss: 0.695 | acc: 75.79%
Epoch 24 | Batch 98/196 | loss: 0.719 | acc: 74.77%
Epoch 24 | Batch 147/196 | loss: 0.699 | acc: 75.75%
Epoch 24 | Batch 196/196 | loss: 0.709 | acc: 75.50%
>> Epoch 24 SUMMARY: Val Acc: 73.45% | Current LR: 0.01
```

```
Epoch 25 | Batch 49/196 | loss: 0.690 | acc: 75.58%
Epoch 25 | Batch 98/196 | loss: 0.695 | acc: 75.41%
Epoch 25 | Batch 147/196 | loss: 0.690 | acc: 75.77%
Epoch 25 | Batch 196/196 | loss: 0.664 | acc: 76.64%
>> Epoch 25 SUMMARY: Val Acc: 76.57% | Current LR: 0.01
Epoch 26 | Batch 49/196 | loss: 0.665 | acc: 77.06%
Epoch 26 | Batch 98/196 | loss: 0.669 | acc: 76.75%
Epoch 26 | Batch 147/196 | loss: 0.674 | acc: 76.31%
Epoch 26 | Batch 196/196 | loss: 0.690 | acc: 75.90%
>> Epoch 26 SUMMARY: Val Acc: 74.41% | Current LR: 0.01
Epoch 27 | Batch 49/196 | loss: 0.653 | acc: 76.90%
Epoch 27 | Batch 98/196 | loss: 0.661 | acc: 77.10%
Epoch 27 | Batch 147/196 | loss: 0.635 | acc: 77.98%
Epoch 27 | Batch 196/196 | loss: 0.660 | acc: 77.12%
>> Epoch 27 SUMMARY: Val Acc: 76.63% | Current LR: 0.01
Epoch 28 | Batch 49/196 | loss: 0.627 | acc: 78.36%
Epoch 28 | Batch 98/196 | loss: 0.633 | acc: 77.81%
Epoch 28 | Batch 147/196 | loss: 0.653 | acc: 77.38%
Epoch 28 | Batch 196/196 | loss: 0.658 | acc: 77.51%
>> Epoch 28 SUMMARY: Val Acc: 75.68% | Current LR: 0.01
Epoch 29 | Batch 49/196 | loss: 0.637 | acc: 78.03%
Epoch 29 | Batch 98/196 | loss: 0.644 | acc: 77.61%
Epoch 29 | Batch 147/196 | loss: 0.644 | acc: 77.60%
Epoch 29 | Batch 196/196 | loss: 0.630 | acc: 77.86%
>> Epoch 29 SUMMARY: Val Acc: 75.86% | Current LR: 0.01
Epoch 30 | Batch 49/196 | loss: 0.601 | acc: 79.12%
Epoch 30 | Batch 98/196 | loss: 0.644 | acc: 77.69%
Epoch 30 | Batch 147/196 | loss: 0.598 | acc: 78.81%
Epoch 30 | Batch 196/196 | loss: 0.627 | acc: 78.33%
>> Epoch 30 SUMMARY: Val Acc: 76.45% | Current LR: 0.01
Epoch 31 | Batch 49/196 | loss: 0.609 | acc: 78.91%
Epoch 31 | Batch 98/196 | loss: 0.599 | acc: 79.21%
Epoch 31 | Batch 147/196 | loss: 0.601 | acc: 79.20%
Epoch 31 | Batch 196/196 | loss: 0.614 | acc: 78.78%
>> Epoch 31 SUMMARY: Val Acc: 76.74% | Current LR: 0.01
Epoch 32 | Batch 49/196 | loss: 0.579 | acc: 79.95%
Epoch 32 | Batch 98/196 | loss: 0.593 | acc: 79.69%
Epoch 32 | Batch 147/196 | loss: 0.598 | acc: 79.00%
Epoch 32 | Batch 196/196 | loss: 0.604 | acc: 79.43%
>> Epoch 32 SUMMARY: Val Acc: 78.11% | Current LR: 0.01
Epoch 33 | Batch 49/196 | loss: 0.563 | acc: 80.74%
Epoch 33 | Batch 98/196 | loss: 0.577 | acc: 79.89%
Epoch 33 | Batch 147/196 | loss: 0.573 | acc: 80.53%
Epoch 33 | Batch 196/196 | loss: 0.603 | acc: 79.25%
>> Epoch 33 SUMMARY: Val Acc: 79.09% | Current LR: 0.01
Epoch 34 | Batch 49/196 | loss: 0.585 | acc: 79.63%
Epoch 34 | Batch 98/196 | loss: 0.571 | acc: 80.21%
Epoch 34 | Batch 147/196 | loss: 0.554 | acc: 80.60%
Epoch 34 | Batch 196/196 | loss: 0.564 | acc: 80.44%
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>> Epoch 34 SUMMARY: Val Acc: 77.63% | Current LR: 0.01
Epoch 35 | Batch 49/196 | loss: 0.564 | acc: 80.61%
Epoch 35 | Batch 98/196 | loss: 0.561 | acc: 80.68%
Epoch 35 | Batch 147/196 | loss: 0.577 | acc: 80.05%
Epoch 35 | Batch 196/196 | loss: 0.551 | acc: 80.76%
>> Epoch 35 SUMMARY: Val Acc: 77.84% | Current LR: 0.01
Epoch 36 | Batch 49/196 | loss: 0.562 | acc: 80.53%
Epoch 36 | Batch 98/196 | loss: 0.550 | acc: 80.98%
Epoch 36 | Batch 147/196 | loss: 0.550 | acc: 81.12%
Epoch 36 | Batch 196/196 | loss: 0.550 | acc: 80.72%
>> Epoch 36 SUMMARY: Val Acc: 79.93% | Current LR: 0.01
Epoch 37 | Batch 49/196 | loss: 0.535 | acc: 81.08%
Epoch 37 | Batch 98/196 | loss: 0.550 | acc: 80.99%
Epoch 37 | Batch 147/196 | loss: 0.525 | acc: 81.71%
Epoch 37 | Batch 196/196 | loss: 0.549 | acc: 80.81%
>> Epoch 37 SUMMARY: Val Acc: 77.89% | Current LR: 0.01
Epoch 38 | Batch 49/196 | loss: 0.526 | acc: 81.94%
Epoch 38 | Batch 98/196 | loss: 0.522 | acc: 81.97%
Epoch 38 | Batch 147/196 | loss: 0.530 | acc: 81.86%
Epoch 38 | Batch 196/196 | loss: 0.541 | acc: 81.61%
>> Epoch 38 SUMMARY: Val Acc: 79.30% | Current LR: 0.01
Epoch 39 | Batch 49/196 | loss: 0.513 | acc: 82.34%
Epoch 39 | Batch 98/196 | loss: 0.529 | acc: 81.74%
Epoch 39 | Batch 147/196 | loss: 0.512 | acc: 82.43%
Epoch 39 | Batch 196/196 | loss: 0.537 | acc: 81.46%
>> Epoch 39 SUMMARY: Val Acc: 79.81% | Current LR: 0.01
Epoch 40 | Batch 49/196 | loss: 0.516 | acc: 82.21%
Epoch 40 | Batch 98/196 | loss: 0.511 | acc: 82.10%
Epoch 40 | Batch 147/196 | loss: 0.497 | acc: 83.25%
Epoch 40 | Batch 196/196 | loss: 0.504 | acc: 82.50%
>> Epoch 40 SUMMARY: Val Acc: 79.85% | Current LR: 0.01
Epoch 41 | Batch 49/196 | loss: 0.496 | acc: 82.92%
Epoch 41 | Batch 98/196 | loss: 0.514 | acc: 82.29%
Epoch 41 | Batch 147/196 | loss: 0.504 | acc: 82.47%
Epoch 41 | Batch 196/196 | loss: 0.496 | acc: 82.88%
>> Epoch 41 SUMMARY: Val Acc: 79.42% | Current LR: 0.01
Epoch 42 | Batch 49/196 | loss: 0.507 | acc: 82.37%
Epoch 42 | Batch 98/196 | loss: 0.502 | acc: 82.57%
Epoch 42 | Batch 147/196 | loss: 0.489 | acc: 83.36%
Epoch 42 | Batch 196/196 | loss: 0.482 | acc: 83.36%
>> Epoch 42 SUMMARY: Val Acc: 80.68% | Current LR: 0.01
Epoch 43 | Batch 49/196 | loss: 0.470 | acc: 83.62%
Epoch 43 | Batch 98/196 | loss: 0.493 | acc: 83.22%
Epoch 43 | Batch 147/196 | loss: 0.474 | acc: 83.32%
Epoch 43 | Batch 196/196 | loss: 0.485 | acc: 83.15%
>> Epoch 43 SUMMARY: Val Acc: 80.61% | Current LR: 0.01
Epoch 44 | Batch 49/196 | loss: 0.480 | acc: 83.67%
Epoch 44 | Batch 98/196 | loss: 0.474 | acc: 83.67%
Epoch 44 | Batch 147/196 | loss: 0.473 | acc: 83.74%
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Epoch 44 | Batch 196/196 | loss: 0.475 | acc: 83.36%
>> Epoch 44 SUMMARY: Val Acc: 79.98% | Current LR: 0.01
Epoch 45 | Batch 49/196 | loss: 0.454 | acc: 84.57%
Epoch 45 | Batch 98/196 | loss: 0.483 | acc: 83.64%
Epoch 45 | Batch 147/196 | loss: 0.476 | acc: 83.39%
Epoch 45 | Batch 196/196 | loss: 0.454 | acc: 84.46%
>> Epoch 45 SUMMARY: Val Acc: 80.31% | Current LR: 0.01
Epoch 46 | Batch 49/196 | loss: 0.451 | acc: 84.10%
Epoch 46 | Batch 98/196 | loss: 0.450 | acc: 84.50%
Epoch 46 | Batch 147/196 | loss: 0.467 | acc: 83.90%
Epoch 46 | Batch 196/196 | loss: 0.450 | acc: 84.23%
>> Epoch 46 SUMMARY: Val Acc: 80.47% | Current LR: 0.01
Epoch 47 | Batch 49/196 | loss: 0.446 | acc: 84.69%
Epoch 47 | Batch 98/196 | loss: 0.455 | acc: 84.31%
Epoch 47 | Batch 147/196 | loss: 0.459 | acc: 84.24%
Epoch 47 | Batch 196/196 | loss: 0.441 | acc: 84.83%
>> Epoch 47 SUMMARY: Val Acc: 78.88% | Current LR: 0.01
Epoch 48 | Batch 49/196 | loss: 0.444 | acc: 85.01%
Epoch 48 | Batch 98/196 | loss: 0.443 | acc: 84.57%
Epoch 48 | Batch 147/196 | loss: 0.455 | acc: 84.57%
Epoch 48 | Batch 196/196 | loss: 0.446 | acc: 84.63%
>> Epoch 48 SUMMARY: Val Acc: 80.97% | Current LR: 0.01
Epoch 49 | Batch 49/196 | loss: 0.435 | acc: 85.03%
Epoch 49 | Batch 98/196 | loss: 0.431 | acc: 85.22%
Epoch 49 | Batch 147/196 | loss: 0.437 | acc: 84.77%
Epoch 49 | Batch 196/196 | loss: 0.437 | acc: 85.04%
>> Epoch 49 SUMMARY: Val Acc: 81.08% | Current LR: 0.01
Epoch 50 | Batch 49/196 | loss: 0.417 | acc: 85.36%
Epoch 50 | Batch 98/196 | loss: 0.427 | acc: 85.15%
Epoch 50 | Batch 147/196 | loss: 0.416 | acc: 85.55%
Epoch 50 | Batch 196/196 | loss: 0.431 | acc: 84.89%
>> Epoch 50 SUMMARY: Val Acc: 81.94% | Current LR: 0.01
Epoch 51 | Batch 49/196 | loss: 0.411 | acc: 86.11%
Epoch 51 | Batch 98/196 | loss: 0.420 | acc: 85.36%
Epoch 51 | Batch 147/196 | loss: 0.417 | acc: 85.79%
Epoch 51 | Batch 196/196 | loss: 0.426 | acc: 85.18%
>> Epoch 51 SUMMARY: Val Acc: 81.82% | Current LR: 0.01
Epoch 52 | Batch 49/196 | loss: 0.401 | acc: 86.33%
Epoch 52 | Batch 98/196 | loss: 0.410 | acc: 85.52%
Epoch 52 | Batch 147/196 | loss: 0.410 | acc: 85.92%
Epoch 52 | Batch 196/196 | loss: 0.412 | acc: 85.97%
>> Epoch 52 SUMMARY: Val Acc: 81.56% | Current LR: 0.01
Epoch 53 | Batch 49/196 | loss: 0.387 | acc: 86.62%
Epoch 53 | Batch 98/196 | loss: 0.408 | acc: 85.87%
Epoch 53 | Batch 147/196 | loss: 0.406 | acc: 85.85%
Epoch 53 | Batch 196/196 | loss: 0.414 | acc: 85.57%
>> Epoch 53 SUMMARY: Val Acc: 81.17% | Current LR: 0.01
Epoch 54 | Batch 49/196 | loss: 0.398 | acc: 86.00%
Epoch 54 | Batch 98/196 | loss: 0.395 | acc: 86.61%
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Epoch 54 | Batch 147/196 | loss: 0.391 | acc: 86.56%
Epoch 54 | Batch 196/196 | loss: 0.393 | acc: 86.21%
>> Epoch 54 SUMMARY: Val Acc: 82.43% | Current LR: 0.01
Epoch 55 | Batch 49/196 | loss: 0.380 | acc: 86.83%
Epoch 55 | Batch 98/196 | loss: 0.395 | acc: 86.59%
Epoch 55 | Batch 147/196 | loss: 0.402 | acc: 85.98%
Epoch 55 | Batch 196/196 | loss: 0.398 | acc: 85.96%
>> Epoch 55 SUMMARY: Val Acc: 80.41% | Current LR: 0.01
Epoch 56 | Batch 49/196 | loss: 0.388 | acc: 86.76%
Epoch 56 | Batch 98/196 | loss: 0.388 | acc: 86.34%
Epoch 56 | Batch 147/196 | loss: 0.370 | acc: 87.11%
Epoch 56 | Batch 196/196 | loss: 0.378 | acc: 86.77%
>> Epoch 56 SUMMARY: Val Acc: 80.71% | Current LR: 0.01
Epoch 57 | Batch 49/196 | loss: 0.373 | acc: 87.02%
Epoch 57 | Batch 98/196 | loss: 0.359 | acc: 87.66%
Epoch 57 | Batch 147/196 | loss: 0.392 | acc: 86.41%
Epoch 57 | Batch 196/196 | loss: 0.390 | acc: 86.80%
>> Epoch 57 SUMMARY: Val Acc: 81.43% | Current LR: 0.01
Epoch 58 | Batch 49/196 | loss: 0.376 | acc: 86.83%
Epoch 58 | Batch 98/196 | loss: 0.368 | acc: 87.41%
Epoch 58 | Batch 147/196 | loss: 0.357 | acc: 87.74%
Epoch 58 | Batch 196/196 | loss: 0.367 | acc: 87.07%
>> Epoch 58 SUMMARY: Val Acc: 82.67% | Current LR: 0.01
Epoch 59 | Batch 49/196 | loss: 0.351 | acc: 88.03%
Epoch 59 | Batch 98/196 | loss: 0.364 | acc: 87.46%
Epoch 59 | Batch 147/196 | loss: 0.377 | acc: 86.87%
Epoch 59 | Batch 196/196 | loss: 0.375 | acc: 87.03%
>> Epoch 59 SUMMARY: Val Acc: 83.05% | Current LR: 0.01
Epoch 60 | Batch 49/196 | loss: 0.357 | acc: 87.79%
Epoch 60 | Batch 98/196 | loss: 0.358 | acc: 87.95%
Epoch 60 | Batch 147/196 | loss: 0.359 | acc: 87.66%
Epoch 60 | Batch 196/196 | loss: 0.352 | acc: 88.01%
>> Epoch 60 SUMMARY: Val Acc: 83.28% | Current LR: 0.01
Epoch 61 | Batch 49/196 | loss: 0.345 | acc: 88.20%
Epoch 61 | Batch 98/196 | loss: 0.348 | acc: 88.10%
Epoch 61 | Batch 147/196 | loss: 0.343 | acc: 88.21%
Epoch 61 | Batch 196/196 | loss: 0.382 | acc: 87.16%
>> Epoch 61 SUMMARY: Val Acc: 82.78% | Current LR: 0.01
Epoch 62 | Batch 49/196 | loss: 0.349 | acc: 87.88%
Epoch 62 | Batch 98/196 | loss: 0.347 | acc: 88.19%
Epoch 62 | Batch 147/196 | loss: 0.343 | acc: 87.97%
Epoch 62 | Batch 196/196 | loss: 0.355 | acc: 87.77%
>> Epoch 62 SUMMARY: Val Acc: 83.10% | Current LR: 0.01
Epoch 63 | Batch 49/196 | loss: 0.347 | acc: 87.88%
Epoch 63 | Batch 98/196 | loss: 0.328 | acc: 88.71%
Epoch 63 | Batch 147/196 | loss: 0.339 | acc: 88.23%
Epoch 63 | Batch 196/196 | loss: 0.330 | acc: 88.75%
>> Epoch 63 SUMMARY: Val Acc: 82.41% | Current LR: 0.01
Epoch 64 | Batch 49/196 | loss: 0.333 | acc: 88.46%
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Epoch 64 | Batch 98/196 | loss: 0.335 | acc: 88.52%
Epoch 64 | Batch 147/196 | loss: 0.333 | acc: 88.54%
Epoch 64 | Batch 196/196 | loss: 0.342 | acc: 88.37%
>> Epoch 64 SUMMARY: Val Acc: 83.71% | Current LR: 0.01
Epoch 65 | Batch 49/196 | loss: 0.309 | acc: 89.16%
Epoch 65 | Batch 98/196 | loss: 0.319 | acc: 88.92%
Epoch 65 | Batch 147/196 | loss: 0.328 | acc: 88.42%
Epoch 65 | Batch 196/196 | loss: 0.348 | acc: 87.94%
>> Epoch 65 SUMMARY: Val Acc: 82.47% | Current LR: 0.01
Epoch 66 | Batch 49/196 | loss: 0.314 | acc: 89.01%
Epoch 66 | Batch 98/196 | loss: 0.326 | acc: 88.66%
Epoch 66 | Batch 147/196 | loss: 0.331 | acc: 88.90%
Epoch 66 | Batch 196/196 | loss: 0.318 | acc: 89.25%
>> Epoch 66 SUMMARY: Val Acc: 82.57% | Current LR: 0.01
Epoch 67 | Batch 49/196 | loss: 0.320 | acc: 89.32%
Epoch 67 | Batch 98/196 | loss: 0.313 | acc: 89.17%
Epoch 67 | Batch 147/196 | loss: 0.315 | acc: 89.22%
Epoch 67 | Batch 196/196 | loss: 0.307 | acc: 89.17%
>> Epoch 67 SUMMARY: Val Acc: 83.15% | Current LR: 0.01
Epoch 68 | Batch 49/196 | loss: 0.314 | acc: 89.23%
Epoch 68 | Batch 98/196 | loss: 0.302 | acc: 89.26%
Epoch 68 | Batch 147/196 | loss: 0.306 | acc: 89.47%
Epoch 68 | Batch 196/196 | loss: 0.319 | acc: 88.87%
>> Epoch 68 SUMMARY: Val Acc: 83.67% | Current LR: 0.01
Epoch 69 | Batch 49/196 | loss: 0.307 | acc: 89.33%
Epoch 69 | Batch 98/196 | loss: 0.293 | acc: 89.72%
Epoch 69 | Batch 147/196 | loss: 0.314 | acc: 89.33%
Epoch 69 | Batch 196/196 | loss: 0.305 | acc: 90.07%
>> Epoch 69 SUMMARY: Val Acc: 83.00% | Current LR: 0.01
Epoch 70 | Batch 49/196 | loss: 0.282 | acc: 90.31%
Epoch 70 | Batch 98/196 | loss: 0.297 | acc: 89.56%
Epoch 70 | Batch 147/196 | loss: 0.308 | acc: 89.21%
Epoch 70 | Batch 196/196 | loss: 0.300 | acc: 89.89%
>> Epoch 70 SUMMARY: Val Acc: 83.39% | Current LR: 0.01
Epoch 71 | Batch 49/196 | loss: 0.286 | acc: 90.08%
Epoch 71 | Batch 98/196 | loss: 0.298 | acc: 89.80%
Epoch 71 | Batch 147/196 | loss: 0.312 | acc: 89.27%
Epoch 71 | Batch 196/196 | loss: 0.302 | acc: 89.62%
>> Epoch 71 SUMMARY: Val Acc: 82.75% | Current LR: 0.01
Epoch 72 | Batch 49/196 | loss: 0.290 | acc: 90.31%
Epoch 72 | Batch 98/196 | loss: 0.296 | acc: 89.71%
Epoch 72 | Batch 147/196 | loss: 0.278 | acc: 90.34%
Epoch 72 | Batch 196/196 | loss: 0.292 | acc: 89.82%
>> Epoch 72 SUMMARY: Val Acc: 81.78% | Current LR: 0.01
Epoch 73 | Batch 49/196 | loss: 0.281 | acc: 90.28%
Epoch 73 | Batch 98/196 | loss: 0.277 | acc: 90.26%
Epoch 73 | Batch 147/196 | loss: 0.277 | acc: 90.12%
Epoch 73 | Batch 196/196 | loss: 0.288 | acc: 89.80%
>> Epoch 73 SUMMARY: Val Acc: 83.76% | Current LR: 0.01
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Epoch 74 | Batch 49/196 | loss: 0.272 | acc: 90.70%
Epoch 74 | Batch 98/196 | loss: 0.273 | acc: 90.34%
Epoch 74 | Batch 147/196 | loss: 0.278 | acc: 90.42%
Epoch 74 | Batch 196/196 | loss: 0.296 | acc: 89.93%
>> Epoch 74 SUMMARY: Val Acc: 84.25% | Current LR: 0.01
Epoch 75 | Batch 49/196 | loss: 0.265 | acc: 91.06%
Epoch 75 | Batch 98/196 | loss: 0.273 | acc: 91.06%
Epoch 75 | Batch 147/196 | loss: 0.279 | acc: 90.38%
Epoch 75 | Batch 196/196 | loss: 0.279 | acc: 90.05%
>> Epoch 75 SUMMARY: Val Acc: 83.74% | Current LR: 0.01
Epoch 76 | Batch 49/196 | loss: 0.260 | acc: 90.94%
Epoch 76 | Batch 98/196 | loss: 0.275 | acc: 90.38%
Epoch 76 | Batch 147/196 | loss: 0.275 | acc: 90.70%
Epoch 76 | Batch 196/196 | loss: 0.269 | acc: 90.77%
>> Epoch 76 SUMMARY: Val Acc: 83.78% | Current LR: 0.01
Epoch 77 | Batch 49/196 | loss: 0.261 | acc: 91.34%
Epoch 77 | Batch 98/196 | loss: 0.265 | acc: 90.86%
Epoch 77 | Batch 147/196 | loss: 0.262 | acc: 90.53%
Epoch 77 | Batch 196/196 | loss: 0.265 | acc: 90.90%
>> Epoch 77 SUMMARY: Val Acc: 84.00% | Current LR: 0.01
Epoch 78 | Batch 49/196 | loss: 0.261 | acc: 90.77%
Epoch 78 | Batch 98/196 | loss: 0.258 | acc: 91.33%
Epoch 78 | Batch 147/196 | loss: 0.252 | acc: 91.37%
Epoch 78 | Batch 196/196 | loss: 0.248 | acc: 91.37%
>> Epoch 78 SUMMARY: Val Acc: 83.38% | Current LR: 0.01
Epoch 79 | Batch 49/196 | loss: 0.237 | acc: 91.84%
Epoch 79 | Batch 98/196 | loss: 0.255 | acc: 91.46%
Epoch 79 | Batch 147/196 | loss: 0.248 | acc: 91.41%
Epoch 79 | Batch 196/196 | loss: 0.255 | acc: 91.34%
>> Epoch 79 SUMMARY: Val Acc: 83.35% | Current LR: 0.01
Epoch 80 | Batch 49/196 | loss: 0.248 | acc: 91.45%
Epoch 80 | Batch 98/196 | loss: 0.251 | acc: 91.26%
Epoch 80 | Batch 147/196 | loss: 0.251 | acc: 91.10%
Epoch 80 | Batch 196/196 | loss: 0.255 | acc: 91.02%
>> Epoch 80 SUMMARY: Val Acc: 83.77% | Current LR: 0.01
Epoch 81 | Batch 49/196 | loss: 0.245 | acc: 91.65%
Epoch 81 | Batch 98/196 | loss: 0.244 | acc: 91.45%
Epoch 81 | Batch 147/196 | loss: 0.255 | acc: 91.25%
Epoch 81 | Batch 196/196 | loss: 0.251 | acc: 91.58%
>> Epoch 81 SUMMARY: Val Acc: 83.60% | Current LR: 0.01
Epoch 82 | Batch 49/196 | loss: 0.235 | acc: 91.80%
Epoch 82 | Batch 98/196 | loss: 0.251 | acc: 91.55%
Epoch 82 | Batch 147/196 | loss: 0.257 | acc: 90.89%
Epoch 82 | Batch 196/196 | loss: 0.239 | acc: 91.47%
>> Epoch 82 SUMMARY: Val Acc: 83.64% | Current LR: 0.01
Epoch 83 | Batch 49/196 | loss: 0.238 | acc: 91.60%
Epoch 83 | Batch 98/196 | loss: 0.240 | acc: 91.81%
Epoch 83 | Batch 147/196 | loss: 0.236 | acc: 91.78%
Epoch 83 | Batch 196/196 | loss: 0.236 | acc: 91.66%
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>> Epoch 83 SUMMARY: Val Acc: 84.12% | Current LR: 0.01
Epoch 84 | Batch 49/196 | loss: 0.237 | acc: 91.76%
Epoch 84 | Batch 98/196 | loss: 0.247 | acc: 91.55%
Epoch 84 | Batch 147/196 | loss: 0.224 | acc: 92.29%
Epoch 84 | Batch 196/196 | loss: 0.236 | acc: 91.97%
>> Epoch 84 SUMMARY: Val Acc: 83.96% | Current LR: 0.01
Epoch 85 | Batch 49/196 | loss: 0.217 | acc: 92.49%
Epoch 85 | Batch 98/196 | loss: 0.235 | acc: 92.06%
Epoch 85 | Batch 147/196 | loss: 0.232 | acc: 91.71%
Epoch 85 | Batch 196/196 | loss: 0.232 | acc: 91.74%
>> Epoch 85 SUMMARY: Val Acc: 82.89% | Current LR: 0.01
Epoch 86 | Batch 49/196 | loss: 0.189 | acc: 93.58%
Epoch 86 | Batch 98/196 | loss: 0.170 | acc: 93.88%
Epoch 86 | Batch 147/196 | loss: 0.168 | acc: 94.30%
Epoch 86 | Batch 196/196 | loss: 0.164 | acc: 94.48%
>> Epoch 86 SUMMARY: Val Acc: 85.81% | Current LR: 0.001
Epoch 87 | Batch 49/196 | loss: 0.154 | acc: 94.99%
Epoch 87 | Batch 98/196 | loss: 0.168 | acc: 94.24%
Epoch 87 | Batch 147/196 | loss: 0.155 | acc: 94.78%
Epoch 87 | Batch 196/196 | loss: 0.155 | acc: 94.46%
>> Epoch 87 SUMMARY: Val Acc: 85.60% | Current LR: 0.001
Epoch 88 | Batch 49/196 | loss: 0.156 | acc: 94.67%
Epoch 88 | Batch 98/196 | loss: 0.151 | acc: 94.89%
Epoch 88 | Batch 147/196 | loss: 0.151 | acc: 94.73%
Epoch 88 | Batch 196/196 | loss: 0.146 | acc: 94.87%
>> Epoch 88 SUMMARY: Val Acc: 85.85% | Current LR: 0.001
Epoch 89 | Batch 49/196 | loss: 0.152 | acc: 94.84%
Epoch 89 | Batch 98/196 | loss: 0.149 | acc: 95.01%
Epoch 89 | Batch 147/196 | loss: 0.149 | acc: 94.97%
Epoch 89 | Batch 196/196 | loss: 0.150 | acc: 94.81%
>> Epoch 89 SUMMARY: Val Acc: 85.98% | Current LR: 0.001
Epoch 90 | Batch 49/196 | loss: 0.141 | acc: 95.23%
Epoch 90 | Batch 98/196 | loss: 0.139 | acc: 95.31%
Epoch 90 | Batch 147/196 | loss: 0.151 | acc: 95.01%
Epoch 90 | Batch 196/196 | loss: 0.141 | acc: 95.16%
>> Epoch 90 SUMMARY: Val Acc: 85.71% | Current LR: 0.001
Epoch 91 | Batch 49/196 | loss: 0.140 | acc: 95.30%
Epoch 91 | Batch 98/196 | loss: 0.146 | acc: 95.01%
Epoch 91 | Batch 147/196 | loss: 0.142 | acc: 95.21%
Epoch 91 | Batch 196/196 | loss: 0.147 | acc: 95.07%
>> Epoch 91 SUMMARY: Val Acc: 85.81% | Current LR: 0.001
Epoch 92 | Batch 49/196 | loss: 0.133 | acc: 95.47%
Epoch 92 | Batch 98/196 | loss: 0.141 | acc: 95.15%
Epoch 92 | Batch 147/196 | loss: 0.145 | acc: 95.08%
Epoch 92 | Batch 196/196 | loss: 0.135 | acc: 95.22%
>> Epoch 92 SUMMARY: Val Acc: 85.40% | Current LR: 0.001
Epoch 93 | Batch 49/196 | loss: 0.137 | acc: 95.27%
Epoch 93 | Batch 98/196 | loss: 0.140 | acc: 95.05%
Epoch 93 | Batch 147/196 | loss: 0.143 | acc: 95.13%
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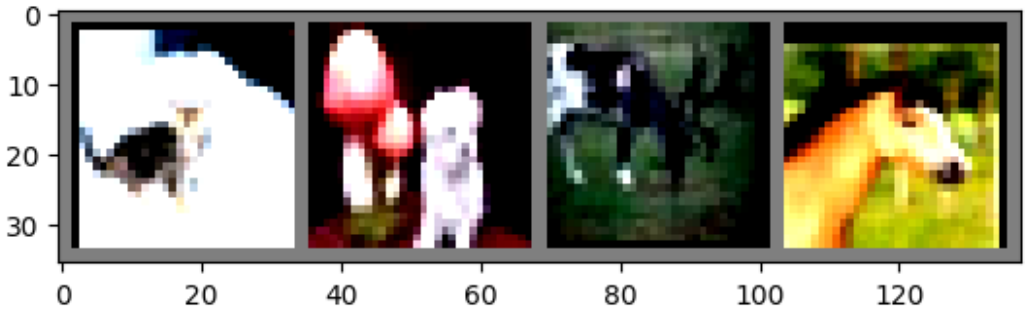
```
Epoch 93 | Batch 196/196 | loss: 0.141 | acc: 95.33%
>> Epoch 93 SUMMARY: Val Acc: 85.23% | Current LR: 0.001
Epoch 94 | Batch 49/196 | loss: 0.140 | acc: 95.22%
Epoch 94 | Batch 98/196 | loss: 0.134 | acc: 95.35%
Epoch 94 | Batch 147/196 | loss: 0.141 | acc: 95.06%
Epoch 94 | Batch 196/196 | loss: 0.133 | acc: 95.33%
>> Epoch 94 SUMMARY: Val Acc: 85.88% | Current LR: 0.001
Epoch 95 | Batch 49/196 | loss: 0.125 | acc: 95.66%
Epoch 95 | Batch 98/196 | loss: 0.137 | acc: 95.54%
Epoch 95 | Batch 147/196 | loss: 0.141 | acc: 95.27%
Epoch 95 | Batch 196/196 | loss: 0.127 | acc: 95.66%
>> Epoch 95 SUMMARY: Val Acc: 85.78% | Current LR: 0.001
Epoch 96 | Batch 49/196 | loss: 0.136 | acc: 95.19%
Epoch 96 | Batch 98/196 | loss: 0.133 | acc: 95.32%
Epoch 96 | Batch 147/196 | loss: 0.130 | acc: 95.62%
Epoch 96 | Batch 196/196 | loss: 0.129 | acc: 95.58%
>> Epoch 96 SUMMARY: Val Acc: 86.02% | Current LR: 0.001
Epoch 97 | Batch 49/196 | loss: 0.136 | acc: 95.20%
Epoch 97 | Batch 98/196 | loss: 0.137 | acc: 95.22%
Epoch 97 | Batch 147/196 | loss: 0.128 | acc: 95.73%
Epoch 97 | Batch 196/196 | loss: 0.132 | acc: 95.42%
>> Epoch 97 SUMMARY: Val Acc: 85.71% | Current LR: 0.001
Epoch 98 | Batch 49/196 | loss: 0.126 | acc: 95.82%
Epoch 98 | Batch 98/196 | loss: 0.131 | acc: 95.47%
Epoch 98 | Batch 147/196 | loss: 0.134 | acc: 95.49%
Epoch 98 | Batch 196/196 | loss: 0.131 | acc: 95.66%
>> Epoch 98 SUMMARY: Val Acc: 86.02% | Current LR: 0.001
Epoch 99 | Batch 49/196 | loss: 0.128 | acc: 95.56%
Epoch 99 | Batch 98/196 | loss: 0.137 | acc: 95.29%
Epoch 99 | Batch 147/196 | loss: 0.128 | acc: 95.59%
Epoch 99 | Batch 196/196 | loss: 0.130 | acc: 95.60%
>> Epoch 99 SUMMARY: Val Acc: 85.74% | Current LR: 0.001
Epoch 100 | Batch 49/196 | loss: 0.132 | acc: 95.33%
Epoch 100 | Batch 98/196 | loss: 0.134 | acc: 95.45%
Epoch 100 | Batch 147/196 | loss: 0.125 | acc: 95.87%
Epoch 100 | Batch 196/196 | loss: 0.121 | acc: 95.92%
>> Epoch 100 SUMMARY: Val Acc: 85.85% | Current LR: 0.001
Finished Training
```

10. Test the network on the test data

```
dataIter = iter(vizLoader)
images, labels = next(dataIter)
# print images
imshow(torchvision.utils.make_grid(images))
print('GroundTruth: ', ' '.join(f'{classes[labels[j]]:5s}' for j in
range(viz_batchSize)))
```

```
Clipping input data to the valid range for imshow with RGB data
([0..1] for floats or [0..255] for integers). Got range [-
0.71453285..1.8378451].
```




```
GroundTruth:  cat   dog   deer  horse
```

```
images, labels = images.to(device), labels.to(device)

outputs = net(images)
_, predicted = torch.max(outputs, 1)

print('Predicted: ', ' '.join(f'{classes[predicted[j].item()]:5s}' for
j in range(len(predicted))))
```

```
Predicted:  cat   dog   dog   horse
```

11. Test the network on the whole dataset

```
test_loss = 0.0
correct = 0
total = 0

with torch.no_grad():
    for data in testLoader:
        images, labels = data[0].to(device), data[1].to(device)

        outputs = net(images)

        # 1. Calculate the loss for this batch
        loss = criterion(outputs, labels)
        test_loss += loss.item()

        # 2. Calculate accuracy as before
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

# Calculate the average loss over all batches
avg_test_loss = test_loss / len(testLoader)
```

```
print(f'Test Loss: {avg_test_loss:.4f}')
print(f'Test Accuracy: {100 * correct / total:.2f}%')
```

```
Test Loss: 0.5339
Test Accuracy: 85.41%
```

12. Check performance within each class

```
# prepare to count predictions for each class
correct_pred = {classname: 0 for classname in classes}
total_pred = {classname: 0 for classname in classes}

# again no gradients needed
with torch.no_grad():
    for data in testLoader:
        images, labels = data[0].to(device), data[1].to(device)
        outputs = net(images)
        _, predictions = torch.max(outputs, 1)
        # collect the correct predictions for each class
        for label, prediction in zip(labels, predictions):
            if label == prediction:
                correct_pred[classes[label]] += 1
            total_pred[classes[label]] += 1

# print accuracy for each class
for classname, correct_count in correct_pred.items():
    accuracy = 100 * float(correct_count) / total_pred[classname]
    print(f'Accuracy for class: {classname:5s} is {accuracy:.1f} %')
```

```
Accuracy for class: plane is 87.1 %
Accuracy for class: car   is 93.3 %
Accuracy for class: bird  is 81.2 %
Accuracy for class: cat   is 72.0 %
Accuracy for class: deer  is 87.0 %
Accuracy for class: dog   is 74.5 %
Accuracy for class: frog  is 90.0 %
Accuracy for class: horse is 88.6 %
Accuracy for class: ship  is 91.9 %
Accuracy for class: truck is 90.8 %
```